
Which genes actually **steer** a T cell — and which just change the most?

An auditable method that separates effect size from functional direction and cross-donor repeatability in CD4+ T-cell Perturb-seq. The deliverable is not a target list — it is a tested scope map that knows when to call PASS, and when to stop.

Abel Costa · clinical research & hematology

A large effect can still be the wrong answer.

Some patients respond to a T-cell therapy, then relapse. To engineer better T cells we must tell apart the genes that **control** a cell's state from those that merely **change** when perturbed. A candidate controller must answer three questions:

REACH

How much did the cell move?

The size of the abnormal result —
necessary, not sufficient.

PRECISION

Which state program moved?

Direction toward activation, Th1/Th2,
memory, Treg or exhaustion.

REPEATABILITY

Did independent units agree?

Different guides and donors must
reproduce the direction.

Magnitude alone is necessary — but not sufficient. Most screens stop at magnitude.

The useful question is: **where did the T cell go?**



M Magnitude — how far the profile moved

C₁ Precision — movement along a frozen state axis

C₂ Repeatability — agreement across biological units

ISCI asks whether C adds information **after** M is known — the pre-specified $M \rightarrow M+C$ test.

PRIMARY RESULT

Direction and repeatability add signal **beyond magnitude.**

+0.357

authoritative full-sample
M → M+C test

95% CI [+0.117, +0.538]

+0.215

leakage-free grouped
out-of-fold gain

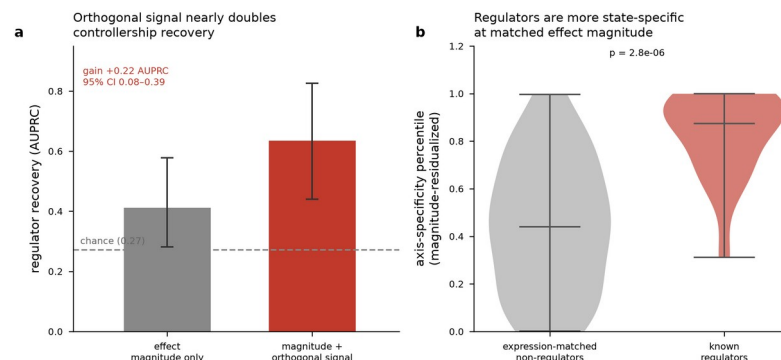
95% CI [+0.074, +0.560] · p = 0.010

0.415 → 0.722

separate descriptive
ranking view

AUPRC on the detectable set

These are two different measurements — not one number shrinking. We report both.



TESTS DESIGNED TO BREAK THE RESULT

The signal is real — and biologically narrow.

We went looking for where it breaks.

External functional-regulator set

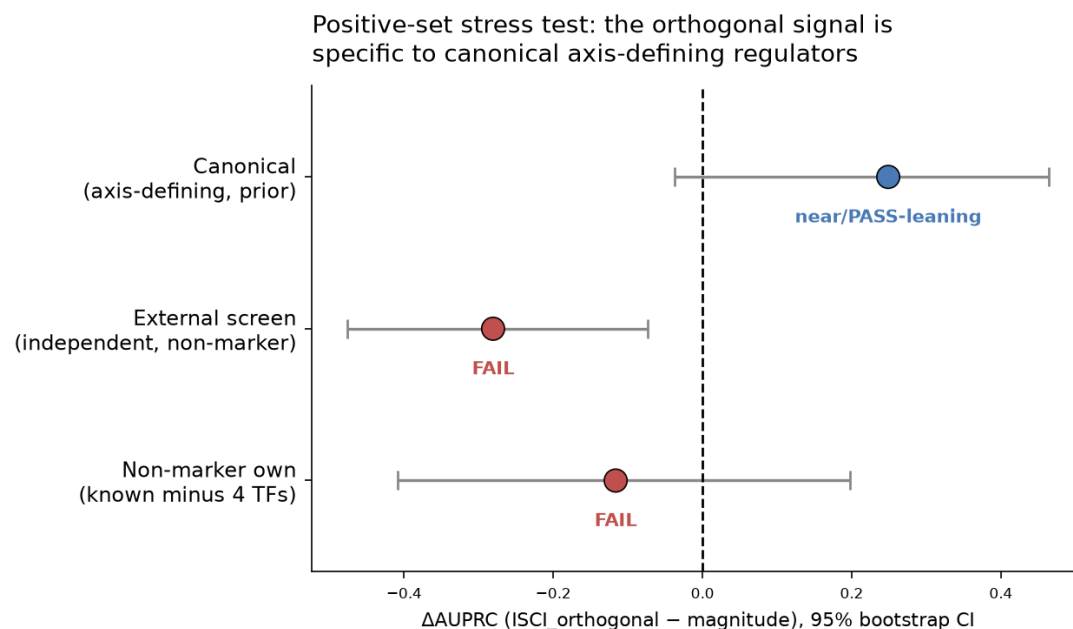
FAIL — ΔAUPRC -0.281 $[-0.476, -0.073]$

Interval sits entirely below zero.

Remove the four master TFs

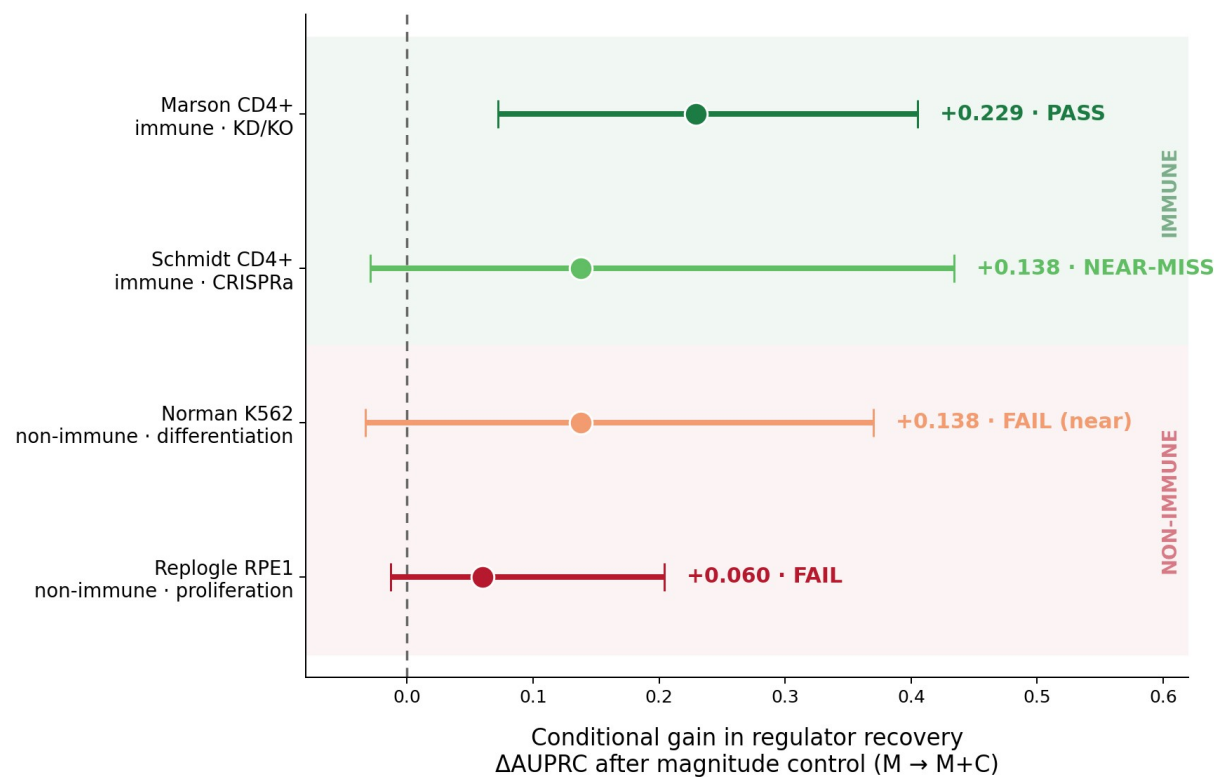
GATA3, TBX21, STAT6, IRF1 → gain ≈ 0 (n=9)

The claim narrows, honestly, to canonical axis-defining regulators in T cells.



External set = 20 non-marker T-cell functional regulators from published CRISPR screens (Shifrut/Schmidt/Belk), expression-matched negatives. FAIL is the honest bound: external functional regulators are magnitude-visible.

The property is **immune-contextual**, not universal.



PASS requires a bootstrap interval excluding zero and the pre-specified conditional gate. A near-miss remains non-PASS.

Marson CD4+

PASS

Schmidt / THP-1

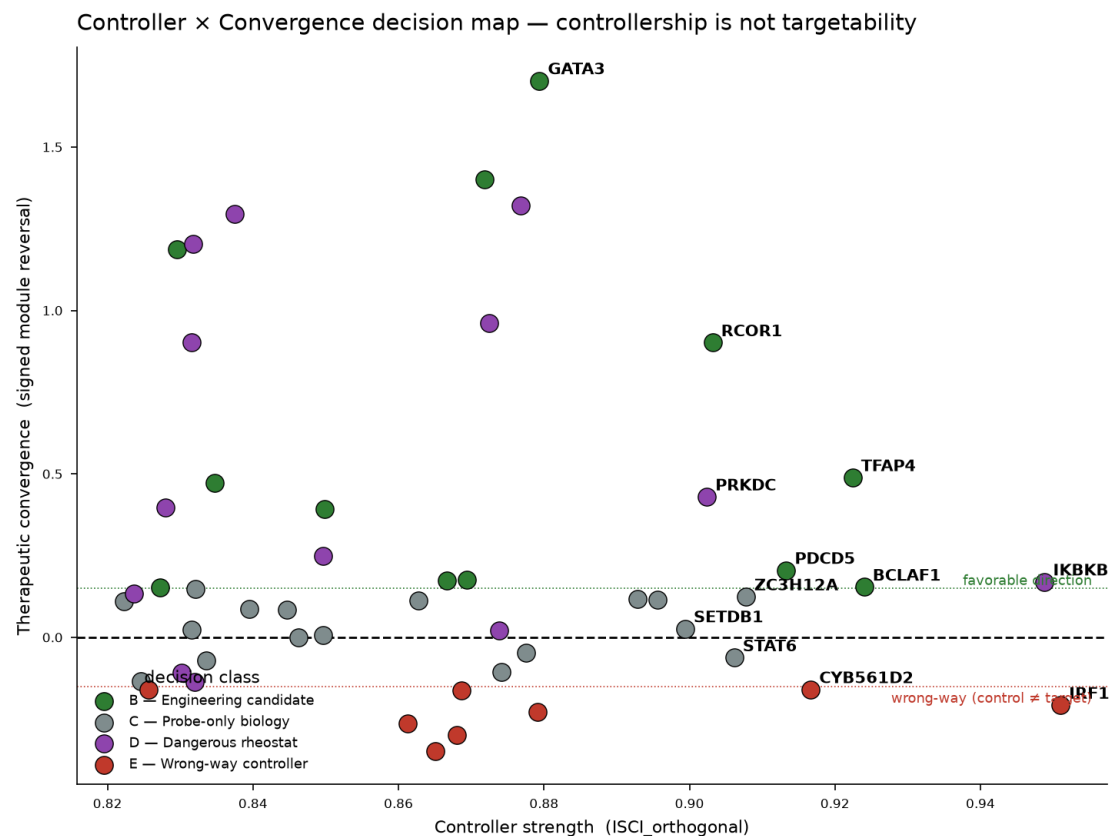
NEAR-MISS

K562 · RPE1

FAIL

A boundary is also a result. We do not use the word *invariant*.

Controller $\neq$ therapeutic target $\neq$ clinical biomarker.



IRF1 ranks #1 as a controller
but points in an unfavorable
intervention direction.

CAR-T response: NULL
study-out AUROC 0.533
CD8 baseline 0.585 beats it
87 patients · 9 studies

*High controllership is not a
promise of clinical benefit.*

The product also knows **when to stop.**

PASS

Pre-specified direction holds and every required gate passes.

FAIL

The claim is testable and a required gate fails.

NULL

The test is valid but adds no supported signal.

NOT-EVALUABLE

A required input is absent; no biology is forced.

Every FAIL and NULL came from a correction loop: Claude challenged the headline, exposed a leakage path, forced the stricter test. **The code owns the numbers; the skepticism is the science.**

A prospective experiment that can falsify the next claim.

DONOR-RESOLVED DESIGN

A frozen ~54-guide CD4 panel across paired rest / stimulation contexts in 8–12 independent donors. The donor is the biological replicate — guides, wells and channels do not inflate n. Promotion requires concordant direction across the pre-declared panel.

FALSIFICATION

The extension fails if:

- residual precision gain below the frozen bound
- donor-bootstrap interval crosses zero
- gene-label / context-exchange $p \geq 0.05$

STOP BEFORE SYNTHESIS

Reference sequence, vector compatibility, on-target activity and off-target risk must pass first. A proposed panel is a design — not an experimentally validated result.

Executable, reusable, **auditable.**

1 · CONTRACTS

Frozen axes, claims, gates, DatasetSpec

2 · PYTHON ENGINE

CLI, adapters, block-bounded H5AD

3 · EVIDENCE

JSON/CSV, figures, hashes, tests, ledger

4 · EXPERIENCE

Notebook, offline HTML, video, deck

Run it on any compatible dataset: **uv run isci pipeline dataset.yaml**

21 / 21 automated release gates passed

MIT code · third-party data retain source terms · CITATION.cff · a tested scope map, not a target list